Annexure 02

Teaching & Examination Scheme and Syllabus

B. Tech Electrical Engineering

3rd and 4th Semester

Batch – EE2020

w.e.f. ACY 2021-2022

Approved by BoS: March 2021

PANDIT DEENDAYAL ENERGY UNIVERSITY, GANDHINAGAR

SCHOOL OF TECHNOLOGY

COURSE STRUCTURE FOR B. TECH. IN ELECTRICAL ENGINEERING													
Semester III			**B. Tech. in Electrical Engineering**										
Sr. No.	**Course/Lab Code**	**Course/Lab Name**	**Teaching Scheme**					**Examination Scheme**					
			L	**T**	**P**	**C**	**Hrs/Wk**	**Theory**			**Practical**		**Total**
								MS	**ES**	**IA**	**LW**	**LE/Viva**	**Marks**
9.	20MA203T	Mathematics-III	3	1	0	4	4	25	50	25	--	--	100
10.	20EE201T	Network Analysis	3	0	0	3	3	25	50	25	--	--	100
11.	20EE202T	Analog & Digital Electronics	3	0	0	3	3	25	50	25	--	--	100
12.	20EE202P	Analog & Digital Electronics - Lab.	0	0	2	1	2	--	--	--	50	50	100
13.	20EE203T	Transformers & Induction Machines	3	0	0	3	3	25	50	25	--	--	100
14.	20EE203P	Transformers & Induction - Machines Lab.	0	0	2	1	2	--	--	--	50	50	100
15.	20EE204T	Open Elective 1	3	0	0	3	3	25	50	25	--	--	100
16.	20HS201P	Communication Skills - II	0	0	2	1	2	--	--	--	50	50	100
		Total	**15**	**1**	**6**	**19**	**22**						**800**

IA- Internal Assessment, MS-Mid Semester; ES – End Semester Exam

Department will offer Open Elective based on availability of expertise/faculty. This open elective will be offered to the students enrolled in B.Tech. programs of School of Technology other than Electrical Engineering.

Student of Electrical Engineering will have a choice for one open elective course from the basket of open electives announced by the other departments of School of Technology/other schools of university.

Subject Code	**Open Elective**
20EE204T	Climate Change: Impact and Solutions

20MA203T					Mathematics - III					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs./Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	1	0	4	4	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept of Fourier series and its application to the solution of partial differential equations.
2. To introduce the Fourier transforms and Z-transforms and understand the application part of it in electrical engineering.
3. To study the first and second order partial differential equations along with their applications in electrical engineering.
4. To study applications of advance Linear Algebra to electrical engineering.

UNIT I: FOURIER SERIES **10 Hrs**

Periodic functions, Euler's formulae, Dirichlet's conditions, expansion of even and odd functions, half range Fourier sine and cosine series, Parseval's formula, complex form of Fourier series.

UNIT II: FOURIER TRANSFORM AND Z-TRANSFORM **10 Hrs**

Fourier Transform: Integral transform, Fourier integral theorem, Fourier sine and cosine integrals, Fourier transforms, Fourier sine and cosine transforms, Properties of Fourier transform, Convolution, Parseval's identity, Relationship between Fourier and Laplace transform.

UNIT III: PARTIAL DIFFERENTIAL EQUATIONS OF FIRST ORDER **10 Hrs**

Formation of Partial Differential Equations (PDEs), Solutions of PDEs of first order, Cauchy problem for first order PDEs, Lagrange's method, Charpit and Jacobi methods for solving first order nonlinear PDEs.

UNIT IV: APPLICATIONS OF ADVANCE LINEAR ALGEBRA TO ELECTRICAL ENGINEERING **10 Hrs**

Introduction to vector space, subspace, basis and dimensions, column space, null space, Field, least squares solutions, orthogonalization, Linear transformation, Symmetric matrices and positive definite matrices, computing exponential of a Matrix, Linearization of non-linear systems using Taylor series expansion

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 – Identify the partial differential equations of first and second order in order to model or understand the Electrical Engineering applications
- CO2 – Understand the techniques of Fourier transforms and Z-transforms to understand the critical mathematical problems.
- CO3 – Apply the methods of Fourier series, Fourier transform and Z-transform in understanding and solving the basic Electrical Engineering problems.
- CO4 – Classify the second order partial differential equations and solve using method of separation of variables.
- CO5 – Appraise the series representation of periodic functions using Fourier series.
- CO6 – Formulate the first order partial differential equations and solve them using various analytical techniques.

TEXT/REFERENCE BOOKS

1. H. K.Dass, **Advanced Engineering Mathematics**, S. Chand & Company Ltd., New Delhi.
2. R.K. Jain & S.R.K. Iyenger, **Advanced Engineering Mathematics**, 3rd Ed., Narosa (2002).
3. E. Kreyszig, **Advanced Engineering Mathematics (Eighth Edition**), John Wiley & Sons.
4. Peter V. O'Neil, **Advanced Engineering Mathematics**, Cengage Learning.
5. K. Sankara Rao, **Introduction to Partial Differential Equations-Third ed.**, PHI Learning.
6. T. Amarnath, **An Elementary Course in Partial Differential Equations**, Narosa Publishing House, New Delhi.

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs.**
Part A: 6 questions 4 marks each	24 Marks (40 min)
Part B: 6 questions 8 marks each	48 Marks (60 min)
Part C: 2 questions 14 marks each	28 Marks (40 min)

20EE201T					NETWORK ANALYSIS					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To provide understanding of electrical networks & its analysis using network theorems, mesh current &nodal voltage method
2. To impart knowledge about AC & DC circuits and determine their behaviour under both steady state & transient condition
3. To understand different properties of signals and linear time-invariant systems
4. To introduce the concept of two port networks and applications of graph theory in network analysis

UNIT I:NETWORK THEOREMS AND GRAPH THEORY **10 Hrs**

Network theorems & circuit analysis techniques: Superposition theorem, Thevenin theorem, Norton theorem, Maximum power transfer theorem, Reciprocity theorem, Compensation theorem, Tellegen's theorem. Analysis with dependent &independent current and voltage sources. Node & Mesh Analysis. Concept of duality & dual networks. Dot convention & Magnetically Coupled Circuit Analysis. **Applications of Graph theory:** Definition of Graph & Various Related Terms, Paths & Circuit Connections, Trees of a Graph, Cut Sets & Tie Sets, Non-Separable Planner & Dual Graphs, Matrices of Oriented Graphs, Properties & Inter Relationships of Incidence, Tie & Cut Set Matrices, Complete Circuit Analysis using Tie Set & Cut Set Matrices

UNIT II: SIGNALS AND LINEAR TIME-INVARIANT SYSTEMS **8 Hrs**

Introduction to signals and systems: Signal properties: periodicity, absolute integrability, determinism & stochastic character. Standard signals: the unit step, the unit impulse, the sinusoid, the complex exponential, Time-limited signals; continuous & discrete time signals, continuous and discrete amplitude signals. System properties: linearity: additivity and homogeneity, shift-invariance, causality, stability, realizability. Examples. **Behaviour of continuous LTI systems:** Impulse response and step response, convolution, correlation, input-output behaviour with aperiodic convergent inputs, cascade interconnections. Characterization of causality and stability of LTI systems. System representation through differential equations. Concepts of complementary function, particular integral, system function. Periodic inputs to an LTI system, notion of a frequency response and its relation to the impulse response.

UNIT III: DC AND AC TRANSIENT ANALYSIS **12 Hrs**

Transient analysis using time domain approach: Solution of first & second order differential equations for Series & parallel R-L, R-C, RL-C circuits, initial & final conditions in network elements, forced and free response, time constants, steady state and transient state response for AC & DC excitation **Transient analysis using Laplace transform approach**: Review of Laplace Transform, Analysis of electrical circuits using Laplace Transform for standard inputs, convolution integral, inverse Laplace transform, transformed network with initial conditions, Transfer function representation, Concept of Poles &Zeroes representation

UNIT IV:TWO PORT NETWORKS AND NETWORK FUNCTIONS **10 Hrs**

Two Port Networks: Two Port Networks, terminal pairs, relationship of two port variables, impedance parameters, admittance parameters, transmission parameters and hybrid parameters, interdependence & relationship, interconnections of two port networks. **Network functions:** Concepts of Complex Frequency, Transform Impedance, Network Functions of 1&2 Port Networks, Concept of Poles &Zeroes, Properties of Driving Point & Transfer Functions, Time Response Stability from Pole-Zero Plot

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Apply different circuit analysis techniques and network theorems for analysing an electrical circuit including a magnetically coupled circuit
CO2 – Apply graph theory to formulate network equations
CO3 – Understand concepts of signals and linear time invariant systems
CO4 – Apply differential equation & Laplace transform approaches to obtain transient response of linear time invariant system
CO5 – Analyse two port networks
CO6 – Understand concepts of network functions

TEXT/REFERENCE BOOKS

1. W.H.Hayt, J.E. Kemmerly, S.M.Durbin, "**Engineering Circuit Analysis**", Tata McGraw-Hill Education India, 8th ed., 2013
2. M.E.VanValkenburg, T.S.Rathore, "**Network Analysis**", Pearson, 3rd ed., 2013
3. F.F.Kuo, "**Network Analysis and Synthesis",** Wiley India Pvt. Ltd, 2nd ed., 2006
4. A.V.Oppenheim, A.V.Willsky, S.H.Nawab,"**Signals and systems**", Pearson, 2nd ed., 2015
5. D.R.Choudhury, "**Networks and Systems**", New Age Intl. Publishers, 2nd ed., 2013

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks
Part B/Question: : 4 Questions, one from each unit, each carrying 10 marks 40 Marks

20EE202T					Analog & Digital Electronics					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To enable the students to understand the fundamentals of analogue integrated circuits and digital electronics.
2. To provide in-depth knowledge about Digital logic ICs, Combinational and Sequential circuits.
3. To emphasize on the significance of low power, small size, reliable, high performance Operational Amplifiers.
4. In addition, the course equips them with the knowledge of basic circuit designing for various applications.

UNIT I INTRODUCTION TO ANALOG DEVICES **10 Hrs.**

Characteristics of diodes, Simple diode circuits: clipping, clamping, rectifiers; Amplifiers: BJT Biasing, Equivalent circuit and Frequency response; Oscillators and Feedback amplifiers; Introduction to Operational Amplifiers (Op-Amp), Ideal Op-Amp, Op-Amp Characteristics, Differential, Inverting & Non-Inverting Amplifiers, Practical Op-Amp (Input Offset Voltage, Input Bias Current, Input Offset Current, Total Output Offset Voltage, Thermal Drift, Common Mode Configuration And CMRR), Op-Amp with Negative Feedback (Voltage-Series & Voltage-Shunt Feedback Amplifier), Frequency Response of Amplifiers.

UNIT II OP-AMP APPLICATIONS **10 Hrs.**

DC & AC Amplifiers, Peaking Amplifier, Summing, Scaling & Averaging Amplifier, Differential Input & Differential Output Amplifier, Integrator & Differentiator, Low Pass Filter, High Pass Filter, Band Reject Filter, Band Pass Filter & All Pass Filter, Basic Comparator, Zero-Crossing Detector, Schmitt Trigger, Window Detector, Voltage Limiters, Voltage to Frequency & Frequency to Voltage Converter, Analog to Digital & Digital to Analog Converters, Voltage Controlled Oscillator, Phase Locked Loop, Fixed & Adjustable Voltage Regulators, 555 Timer as Astable, Bi-Stable & Mono-Stable Multi-Vibrators.

UNIT III BOOLEAN ALGEBRA, LOGIC GATES & COMBINATIONAL CIRCUITS **10 Hrs.**

Binary Arithmetic, Binary Codes, Binary Logic, Basic Theorems & Properties of Boolean Algebra, Boolean Functions, Canonical & Standard Forms, Digital Logic Gates & Their Properties, K-Map Method, Four/Five Variable Map, POS & SOP Simplification, Don't Care Conditions, NAND & NOR Implementation, Exclusive OR Functions.

COMBINATIONAL LOGIC: Combinational Circuit – Analysis & Design, Binary Adder & Subtractor, Decimal Adder, Binary Multiplier, Decoder, Encoder, Multiplexer & De-Multiplexers.

UNIT IV SYNCHRONOUS SEQUENTIAL LOGIC **10 Hrs.**

Sequential Circuits, Latches, Flip-Flops & Excitation Tables, Analysis of Clocked Sequential Circuits, Design of Sequential Circuits, Registers, Shift registers, Ripple Counter, Synchronous Counters, Ring/Johnson Counter.

TOTAL HOURS 40 Hrs.

COURSE OUTCOMES

CO1 – Analyse basic op-amp circuits & feedback amplifiers
CO2 – Understand basic applications of op-amp for analogy & digital circuits
CO3 – Apply algebraic & mapping techniques to minimise the hardware implementation
CO4 – Design, analyse and implement sequential circuits with timing diagram
CO5 – Develop analogue & digital logic to solve real time engineering problems
CO6 – Demonstrate the ability to design practical circuits.

TEXT/REFERENCE BOOKS

1. R. A. Gaikwad, "**Operational Amplifiers and Linear Integrated Circuits**", Prentice Hall of India.
2. Morris Mano, "**Digital Design**", Prentice Hall of India.
3. Donald Leach, Albert Malvino, and GoutamSaha, "**Digital Principles and Applications**", Tata McGraw-Hill.
4. Anand Kumar, "**Switching Theory and Logic Design**", Prentice Hail of India

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100	**Exam Duration: 3 Hrs**
Part A/Question: 4 Questions, one from each unit, each carrying 15 marks	60 Marks
Part B/Question: 4 Questions, one from each unit, each carrying 10 marks	40 Marks

20EE202P					Analog & Digital Electronics – Lab.					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

Course Objective

1. To illustrate different electronic circuits and their practical applications.
2. To test and evaluate the performance of different analogue and digital circuits.
3. To simulate various analogue and digital circuits using software based tools and draw parallel between the simulations and hardware circuits performance.

List of Experiments:

1 Observing Open-Loop Gain of an Operational Amplifier as a function of frequency and measuring Common Mode Rejection Ratio.
2 To Study Operational Amplifier as Inverting and Non Inverting Amplifier, Voltage Comparator, Integrator and Differentiator.
3 To study Active Low Pass Filters, Active High Pass Filter, Active Band Pass Filter using Op-Amp.
4 To study Astable, Mono-stable and Free Running Multi-vibrators using IC 555.
5 To Study Laws and Theorems of Boolean algebra.
6 Study of Logic Gates and Verification of truth tables of Logic gates Using Universal gates (NAND and NOR gates).
7 To verify (A) Truth Table of Binary Half Adder, (B) Truth Table of Binary Full Adder (using two half adders) and (C) Truth Table of Binary Half Subtractor
8 Study of Parity Generator/Checker.
9 To study and verify (A)Gray to Binary Converter, (B) Binary to Gray Converter, and (C) BCD to Excess-3 code conversion circuit.
10 To verify the truth table for 4 To 1 Line Multiplexer and 1 To 4 Line De-Multiplexer
11 To study and verify the Truth Table of 8-to-3 Line Encoder and 3-to-8 Line Decoder.
12 Study of Various types of Flip Flop.
13 Study of Left Right and Programmable Shift Register.
   A. Study of 4-bit serial in serial out shift register.
   B. Study of 4-bit serial in parallel out shift register.
   C. Study of 4-bit parallel in-serial out shift register.
   D. Study of 4-bit parallel in parallel out shift register.
14 Study of 4 Bit Counters (Synchronous and Asynchronous).
   A. Study of 4-bit Synchronous Binary up Counter.
   B. Study of 4-bit Asynchronous Binary up/down counter.
15 Study and verification of Diode VI Characteristics.
16 Application of Diode as Clipper and Clamper Circuits.
17 Conversion of AC to DC using Diode Rectifier Circuits.
18 Study and verification of Transistor in CB and CE configuration (Input and Output Characteristics).
19 To study importance of biasing of Transistor.
20 Study and experimental verification of Hartley Oscillator and Colpitts Oscillator.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Understand basic operation of integrated circuits of analog and digital electronic devices such op-amps, logic gates etc.
CO2 – Develop basic analogue & digital circuits for real-time applications such as ac-dc converter, logic gates etc.
CO3 – Learn software tools available for simulating analogue & digital circuits
CO4 – Select appropriate IC modules for to build hardware for given application
CO5 – Verify and compare the performance through hardware operations and simulation results
CO6 - Demonstrate the ability to design practical circuits.
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END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20EE203T					Transformers & Induction Machines					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand basic principles, working, construction, classification and operating characteristics of single phase as well as three phase AC machines such as Transformers and Induction machines
2. To evaluate and analyse the performance of Transformers & Induction machines
3. To enable students to identify and solve the AC machines related problems in Transformers & Induction motor
4. To impart sound knowledge about the different applications of Transformers & Induction machines

UNIT I Single Phase Transformers **10 Hrs**

Basics of Magnetics Circuits and Electromechanical Energy Conversion, Types, Working Principle, Construction, EMF Equation, Transformer on No-Load & ON Load, Ideal Transformer, Actual Transformer, Vector Diagrams, Equivalent Resistance & Reactance, Equivalent Circuits, Losses, Efficiency & Voltage Regulation, All Day Efficiency, Direct Load Test, O.C. &S.C. Tests, Sumpner's Test, Polarity Test, Parallel Operation & Load Sharing, per unit impedance, Auto Transformers, Applications, Testing and Trouble Shooting of Single Phase Transformer

UNIT II Three Phase Transformers **10 Hrs**

Construction, Types, Vector Groups, Connections (Including Open Delta), Parallel Operation of 3-Phase Transformers, Scott Connection, Three Winding Transformer, Tertiary Winding, Voltage Regulation & Tap Changers, Magnetizing Inrush, Harmonics in Transformer, Cooling Methods, Protective & Safety Devices Fitted on Transformers, Power & Distribution Transformer, Testing and Trouble Shooting of Three Phase Transformer

UNIT III Three Phase Induction Machine **15 Hrs**

Classification of AC Motors, Construction & Types, Working Principle, Production of Rotating Magnetic Field, Synchronous Speed, Slip, Frequency of Rotor Current, rotor parameters under standstill and running condition, Torque, Torque/Slip Characteristics, Power Stages, No Load & Blocked Rotor Tests, Phasor Diagram, Equivalent Circuit, Circle Diagram & Determination of Performance Parameters, Deep Bar & Double Cage Induction Motors, Starters: Automatic and Semi - Automatic starters, Methods of Speed Control, Harmonics & its Effects, Cogging & Crawling, Induction Generators: Working, Effect of Capacitor, Voltage Build up in Self-Excited Generators, Application, Testing and Trouble Shooting of Three Phase Induction Machine

UNIT IV Single Phase Induction Motor **05 Hrs**

Types of Single Phase Motors, Double Revolving Field Theory, Methods of Starting, Equivalent Circuit, No Load & Blocked Rotor Tests, Determination of Equivalent Circuit Parameters, Performance Calculations.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 – Understand the construction, principle of operation and working of different kinds of single phase and three phase AC machines such as Transformers and Induction machines.
- CO2 – Analyze the performance of single phase and three phase transformers.
- CO3 – Understand the importance and application of three phase transformer connections.
- CO4 – Evaluate the performance of three phase induction machines by graphical and analytical methods.
- CO5 – Compare the different methods of starting and speed control of induction motor.
- CO6 – Determine the parameters of equivalent circuit of single phase motor and calculate its performance.

TEXT/REFERENCE BOOKS

1. B.L. Theraja& A.K. Theraja, **A text book of Electrical Technology Volume II-AC & DC Machines**, S. Chand & Co.
2. J.B.Gupta, **Theory and performance of electrical machines**, Katson Publication, S.K.Kataria& Sons.
3. Ashfaq Hussain, **Electric Machines**, Dhanpat Rai and Co.
4. D.P. Kothari and I. J. Nagrath, **Electric Machines**, Tata McGraw Hill
5. P. S. Bimbhra, **Electrical Machinery**, Khanna Publishers
6. M. G. Say, **Alternating Current Machines**, Pitman and Sons
7. J. G. Jamnani, **Electrical Machines**, Mahajan Publishing House

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 15 marks 60 Marks

Part B/Question: 4 Questions, one from unit I, one form unit II and 2 from unit III, each carrying 10 marks 40 Marks

20EE203P					Transformers & Induction Machines – Lab.					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To impart the practical knowledge of construction and working of electrical machines like Transformers and Induction machines
2. To perform different types of tests on electrical machines like Transformers and Induction machines
3. To determine the performance characteristics of rotating electrical machines like Transformers and Induction machines
4. To understand starting and speed control methods of Induction Motors.

List of Experiments:

1. To perform polarity and voltage ratio test on Single Phase Transformer.
2. To perform load test on Single Phase Transformer.
3. To perform open circuit and short circuit test on Single Phase Transformer.
4. To determine the parameters of an equivalent circuit of a Single Phase Transformer.
5. To perform Sumpner's Test on Single Phase Transformer
6. To perform parallel operation of Single Phase Transformers.
7. To perform load test on Three Phase Transformer to find out efficiency and regulation.
8. Parallel operation of two Three Phase Transformers.
9. To understand and verify the different vector groups in Three Phase Transformer connections.
10. To obtain 2-phase supply from 3-phase supply using Scott connection.
11. To understand the construction and operation of Three Phase Induction Motor by working cut section.
12. To determine equivalent circuit parameters of Three Phase Induction Motor by No load and Blocked rotor test.
13. To determine performance characteristics of Three Phase Induction Motor using circle diagram.
14. To perform load test on a Three Phase Induction Motor.
15. To study different types of starters used for Three Phase Induction Motor.
16. To determine equivalent circuit parameters of Single Phase Induction Motor by No load and Blocked rotor test.
17. To study starting methods of a Single Phase Induction Motor.
18. To perform load test on Single Phase Induction Motor.

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Demonstrate the construction, working and operation of AC machines like Transformers and Induction Machine
CO2 – Verify and Analyze the performance characteristics of AC machines like Transformers and Induction Machine by conducting different tests
CO3 – Estimate the performance of Induction Machines by using circle diagram
CO4 – Understand the different configuration of Three Phase Transformers
CO5 – Understand the speed control of Induction Motors
CO6 – Understand the parallel operation of two Transformers

END SEMESTER LABORATORY EXAMINATION PATTERN

Max. Marks: 100

Continuous evaluation	50 marks
End semester examination and Viva-voce	50 marks

20HS201P					Communication Skills – II					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	50	50	100

COURSE OBJECTIVES

1. To understand communication and its process and effect on giving and receiving information.
2. To learn and apply communication skills in different public and interpersonal contexts.
3. To develop analytical, research, and organizational skills through communication skills for a fulfilling career.

UNIT I: **7 Hrs**

- Technical Writing
  - ✓ Report Writing
  - ✓ Creating Lab Journals and Manuals
- Portfolio of Critical Writing and Creative Writing
  - ✓ Essay, Story-writing, etc.

UNIT II: **7 Hrs**

- Summarizing
- Writing Reviews (Books/Articles/Movies/websites)
- Reading Skills (Advanced)

UNIT III: **7 Hrs**

- Digital Literacy
  - ✓ Emails
  - ✓ Creating e-content
  - ✓ Editing and proofreading online
  - ✓ Using grammar and spell check software
  - ✓ Using plagiarism checkers

UNIT IV: **9 Hrs**

- Group Discussion
- Resume Writing
- Interview Skills

TOTAL HOURS 30 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

CO1 – Apply current technology for effective communication leading to better dissemination of knowledge &expertise.

CO2 – Demonstrate relevant knowledge of communication skills in different settings to cater to different purposes & audiences.

CO3 – A sound understanding of communication theory, practice and application to optimize career opportunities.

CO4 – Dynamic communication skills to build and maintain robust and effective professional relationships.

CO5 – Augmented communication skills to prepare and present messages, reports and documents in intent and to integrate different sources of information and knowledge.

CO6 – Monitoring and critical reflection on communication skills for the adoption of appropriate strategy required in achieving the desired outcomes

TEXT/REFERENCE BOOKS

1. Harmer, Jeremy. **The Practice of English Language Teaching**. Harlow: Pearson Longman, 2007.
2. Kaul, Asha. **Business Communication**. Delhi: Prentice-Hall of India, 2006.
3. Maley, A. '**Literature in the Language Classroom'**, The Cambridge Guide to Teaching ESOL, Cambridge University Press.
4. Richards, Jack C., and Willy A. Renandya, eds. **Methodology in Language Teaching: An Anthology of Current Practice**. Cambridge University Press, 2002.
5. Sharma, Sangeeta and Binod Mishra. **Communication Skills for Engineers and Scientists**. New Delhi: PHI Learning Pvt. Ltd., 2009.

Assessment Tool	Marks	Assignments
Lab Work	50	• Essay/Journal Writing – 10 • Report Writing – 10 • Creating e-content – 10 • Blog Writing – 10 • Review Writing – 10
Lab Exam/Viva	50	• Mock Interview – 15 • Group Discussion – 15 • Cover Letter/Curriculum – 20

20EE204T					Climate Change: Impact and Solutions					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
3	0	0	3	3	25	50	25	--	--	100

COURSE OBJECTIVES

1. To understand the concept of climate change and current status.
2. To understand the concept present generation of renewable energy sources and distributed generation of power sources.
3. To acquire knowledge of basic types of available energy sources.
4. To know the basics about energy conservation and its methodology.

10 Hrs

UNIT I: INTRODUCTION TO THE CLIMATE CHANGE

Earth's Climate, The Greenhouse Effect, The Carbon Cycle, Natural Climate Change, The Human Impact on climate change, Human and biological waste, Heat-waves and droughts, Melting of ice caps and Warming oceans, Losses of the species, Impact on ecological system, Societal impact, Documentaries reported on energy crises and their solutions, Carbon foot prints, Future prediction and summary.

UNIT II: THE ENERGY ENGINEERING **07 Hrs**

Basics of energy, unit conversions, Trends of energy consumption, Scenario of energy consumption in case of developed and developing country, Indian energy scenario, Environmental concern importance of renewable energy, Consumption from Fossil fuel, availability and limitations, Types of Renewable Energy Sources, Need to develop new energy sources- energy.

UNIT III: ENERGY EFFICIENCY AND ENERGY CONSERVATION **08 Hrs**

Concept of energy efficiency, Green transportation, Energy conservation and its importance, energy strategy for the future. Energy Conservation Act 2001 and related policies: Energy conservation Act 2001, Methods of Energy conservation.

UNIT IV: RENEWABLE ENERGY SOURCES AT A GLANCE **15 Hrs**

Solar Energy: Concept of Harnessing Solar Energy, Solar Water Heating, Solar Cooling, Solar Thermal Electricity, Solar Collectors, Introduction to Solar Photovoltaic.

Wind Energy: Concept, Motion of wind, Conversion of wind power, Wind turbine dynamics.

Tidal Power: Concept, Tidal Turbine, Types of Tidal Power Plant.

Biomass: Definition and types of biogas plants and its design for small community.

Geothermal: Definition and Overview of geysers and geothermal power station.

TOTAL HOURS 40 Hrs

COURSE OUTCOMES

On completion of the course, student will be able to:

- CO1 – Remembering basic terms of climate change, non-renewable and renewable sources.
- CO2 – Illustrate the basic terminologies related with energy engineering.
- CO3 – Develop the concept of energy conservation and energy efficiency.
- CO4 – Understand the energy scenario and significance of power generation from renewable energy sources.
- CO5 – Understand the types of alternative energy sources.
- CO6 – Distinguish the various renewable power generation sources

TEXT/REFERENCE BOOKS

1. J. Woodward, "**Eyewitness Climate Change**", DK Publishing.
2. Rao &Parulekar, **Energy Technology** - Khanna Publications, New Delhi, 2007.
3. Sawhney G. S., "**Non-conventional energy sources**", PHI Learning Pvt. Ltd.
4. G. D. Rai, "**Renewable Energy Sources**", Khanna Publishers
5. The Energy Conservation Act, 2001, Ministry of Law, Justice And Company Affairs, Legislative Department.

WEBPAGES

1. https://beeindia.gov.in/
2. https://mnre.gov.in/

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A/Question: 4 Questions, one from each unit, each carrying 20 marks 80 Marks

Part B/Question: 2 Questions, one from unit 1 and one from unit IV, each carrying 10 marks 20 Marks